

Database Management Systems

CPS 3740

Lecture 16

Computer Science, Kean University

Dr. Ching-yu (Austin) Huang

Instructor Paolien Wang

Outline

- Refresh INSERT, UPDATE, DELETE operations
- INSERT, UPDATE, DELETE operations with auto increment and foreign keys in PHP programs
- Project
- Last withdraw date soon

Database Operations

- Operations of the **database data** (**NOT structure**) can be categorized into **retrievals** and **updates**
- Retrieval: **SELECT** values from table(s)
- An example to retrieve data in MySQL
SELECT * FROM Vendors WHERE name='John Smith';
- Basic operations that **change** the states of tables in the database:
 - INSERT
 - DELETE
 - UPDATE (or Modify)

The INSERT Operation

- Provides a list of attribute values for a new tuple that is to be inserted into a table
- Can violate the constraints
- If an insertion violates one or more constraints
 - Default option is to reject the insertion
 - Cause errors
- INSERT with a foreign key
 - More discussions later

INSERT Operation

- **INSERT** statement: SQL command for adding data
- Example 1, add a record with column names:
 - > Insert into Vendor (id,Name,Address,TEL,Contact_person) Values (1,'ABC','100 Abc Ave.\, Union\, NJ\, 07083', '908-737-1000', 'John Smith');
- Example 2, add a record without column names:
 - > Insert into Vendor Values (2, 'OPQ', '200 Opq St.\, Newark\, NJ 07083', '908-737-2000', 'Joe Lee');
- Note: a “\” must be before every comma in the values
- Use MYSQL “desc” command to see table information
 - > desc Vendors;

The UPDATE Operation

- Need to specify columns and the where condition
 - Select the tuple (or tuples) to be modified
 - Ex. Update Students set name='test' where id=3;
- If an attribute is neither part of a primary nor a foreign key
 - Usually causes no problems
- Updating a primary/foreign key
 - Sequence: Similar issues as with Insert/Delete

UPDATE Operation

- **UPDATE:** SQL command for changing data
- Example 1: update one column
- Update Vendors set Name='ABC Inc.' where id=1;
- Example 2: update two columns
- Update Vendors set Name='ABC Inc.', TEL='908-737-1100' where id=1;
- **How update a text with quotes (single or double)?**
- **Be careful!!! (without condition)**
Update Vendors set Name= 'ABC Inc.';

DELETE Operation

- **DELETE**: SQL command for removing records
- Example: delete records with the conditions
 - > delete from Vendors where id=1;
- **Be careful!!! (without condition)**
 - > delete from Vendors;
- DELETE with a foreign key

HTML input and INSERT in PHP

- HTML inputs tips
 - User might fill some fields → NULL, NOT empty
 - Auto increment field should not be entered.
- Receive HTML CGI inputs in PHP
 - PHP variable default values need to match DB default.
 - \$cars, \$comment
- PHP variable values in the INSERT statement
 - `$query = 'INSERT INTO Product_xxxx values ($var1,..... ' ;`
- Execute the INSERT statement inside the PHP
- Always verify the return status.

Refresh CREATE/Drop table, INSERT, UPDATE, DELETE

- Create a table
 - Exercise using table names: MyTable_xxxx
 - Create table MyTable_xxxx (id int auto_increment primary key, name varchar(255));
 - Get permissions on the class website
- Refresh
 - INSERT
 - INSERT ... SELECT ...
 - UPDATE
 - DELETE
- Drop a table
 - Drop table Test_xxxx;

Refresh Auto Increment

- Refresh – what is auto_increment?
- Create table MyTable_xxxx (id int auto_increment primary key, name varchar(255));
- Insert two records
 - INSERT INTO MyTable_xxxx (name) values ('Austin'), ('Mary');
- Can we insert a record with id?
 - INSERT INTO MyTable_xxxx values (5, 'Austin');
- What happens to the id? What is the next value?

Case study:

Real example - mini Amazon

- A customer can order several products at one time
 - Different quantity
- A product can be ordered by several customers
 - Different date
 - Different quantity
- Define the basic tables with minimal attributes
 - Customers (id, name, balance, zipcode)
 - Products (id, name, price, qty)
 - Orders (order_id, customer_id, order_date)
 - Product_Order (order_id, product_id, quantity)
- What is the data flow if we place an order with two items?

Case study (cont.): Tables

- Customers_demo

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	
name	varchar(255)	NO		NULL	
balance	float	NO		NULL	
zipcode	varchar(12)	NO	MUL	NULL	

- Orders_demo

Field	Type	Null	Key	Default	Extra
order_id	int(11)	NO	PRI	NULL	auto_increment
customer_id	int(11)	YES	MUL	NULL	
order_date	date	YES		NULL	

- Products_demo

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	
Name	varchar(20)	YES		NULL	
Price	float	YES		NULL	
qty	int(11)	NO		NULL	

- Product_Order_demo

Field	Type	Null	Key	Default	Extra
order_id	int(11)	YES	MUL	NULL	
product_id	int(11)	YES	MUL	NULL	
quantity	int(3)	NO		NULL	

Case study (cont.):

Data flow & SQL for the ordering

- Need to have customer and products records. (**INSERT**)
- **Start the transaction (general concept, our project does not use transaction)**
- After submitting the order, check if the customer balance is enough for all items. (**UPDATE** balance)
- If not enough, reject the order and **ABORT** the transaction. Nothing should be changed.
- If yes, check each item has enough quantity (**UPDATE** qty for each item in the Products table)
 - If one of the item does not have enough quantity, **ABORT** the transaction. Nothing should be changed.
- If all products have enough quantity, do these in sequence
 - **INSERT** the order into **Orders** table where **order_id** is **auto_increment**.
 - Get the order_id for the inserted order record with customer_id (cookie).
 - **INSERT** the order details into **Product_Order** table with the order_id, product_id, and order quantity for each item in a while loop.
 - **COMMIT** the transaction. Save changes into disk permanently.

Case study (cont.):

MySQL in PHP : INSERT + UPDATE

- INSERT: => check result status

```
$query="INSERT into Product_Order_demo  
(order_id, product_id, quantity) VALUE  
($order_id," . $product_id . "," . $order_quantity . ")";
```

```
$result = mysqli_query($con,$query);
```

- Update: => check affected_rows status

```
$query="UPDATE Products_demo  
SET qty = qty -" . $order_quantity . "  
WHERE product_id=" . $product_id;
```

```
$result = mysqli_query($con,$query);
```

```
$affected_row = mysqli_affected_rows($con); // see next slide
```

```
# check affected_row  
if($affected_row > 0 ) {  
    echo " $affected_row rows are updated!\n";  
}
```

PHP mysqli_affected_rows()

- https://www.w3schools.com/php/func_mysqli_affected_rows.asp

1. Perform queries and print out affected rows: **select**
query= "**SELECT** * FROM \$tabeName";
mysqli_query(\$con,query);
echo "Affected rows: " . **mysqli_affected_rows(\$con)**;
2. Perform queries and print out affected rows: **delete**
query= "**DELETE** FROM \$tabeName where name='Johnny'";
mysqli_query(\$con,query);
echo "Affected rows: " . **mysqli_affected_rows(\$con)**;

Case study (cont.): Handle auto increment id

- Get the product id in PHP
 - Product id should be assigned by the database.
 - Same as Student id, employee id, ... etc
 - The product id is automatically increased.
- If we order 2 products now on the mini Amazon
 - 1 laptop, \$1000 each
 - 2 pens, \$2 each
- The order_id is auto_increment used internally
- How can we get the order_id in PHP?
 - We need to insert the information into Product_Order table

Case study (cont.):

PHP mysqli_insert_id() – Auto increment

- Get the **order_id (auto_increment)** for an **insert** statement

```
$query="INSERT INTO Orders_demo  
VALUES ($customer_id,CURDATE());"
```

```
$result = mysqli_query($con,$query);
```

```
$order_id=mysqli_insert_id($con); // google this function in w3schools
```

```
if ($order_id==0) { echo "Insert error! $query<br>\n"; }
```

About `mysqli_free_result()`

- If the `$result` is empty or null, we should not have the following statement because the `$result` has nothing to free.
 - `mysqli_free_result($result);`
- Otherwise, there will be a warning message.
 - **Warning:** `mysqli_free_result()` expects parameter 1 to be `mysqli_result`,
- For example, these SQL commands
 - `INSERT`, `UPDATE`, `DELETE`

Class exercise in class DB:

Create a table with primary key

```
CREATE TABLE MyTable_xxxx(  
  id int auto_increment,  
  name varchar(20),  
  Primary key(id)  
);
```

- Reference

- https://www.w3schools.com/sql/sql_foreignkey.asp

Class exercise - INSERT and PHP

- PHP Get ID of Last Inserted Record
 - https://www.w3schools.com/php/php_mysql_insert_lastid.asp
- If we have a table MyTable_xxxx (id, name) in **class_database** (eg:TECH3740_2021F), where id is auto increment.

```
$sql = "INSERT INTO <class_database>.MyTable_xxxx (name) VALUES ('John')";
```

```
try {  
    mysqli_query($con, $sql);  
    $last_id=mysqli_insert_id($con);  
    echo "Inserted successfully. Last inserted ID is: ". $last_id ."\n";  
}  
catch (Exception $e) {  
    echo "Something wrong....\n";  
    echo "<br>" . mysqli_error($con);  
}
```

Class exercise - UPDATE and PHP

- Based on these codes, can we tell if the record has been successfully updated? **// check next slide**

```
$query = "update MyTable_xxxx set name='Johnny' where id=1";  
try {  
    mysqli_query($con, $query) ;  
    echo "Successfully changed name to 'Johnny' for id=1. ";  
}  
catch (Exception $e) {  
    echo "Error: ". $query . "<br>" . mysqli_error($con);  
}
```

MySQL in PHP – UPDATE :

use `mysqli_affected_rows` function

- Get number of affected **rows** for an **update, delete** statements

- update.php

```
include "dbconfig.php";
$con = mysqli_connect($host, $user, $pass, $dbname);
$query = "update $dbname.MyTable_xxxxx set name='Johnny' where id=1";
echo "<br>$query\n";
try {
    $result = mysqli_query($con, $query);
    $affected_row = mysqli_affected_rows($con);

    # check affected_row
    if($affected_row > 0 ) {
        echo "Update ... successfully: update $affected_row record!\n";
    }
    else {
        echo "Update error or no such record! <br> query: $query <br> ";
    }
}
catch (Exception $e) {
    echo "Error: ". $query . "<br>" . mysqli_error($con);
}
mysqli_close($con);
```

Class exercise - DELETE and PHP

- Based on the following codes, can we tell if the record has been successfully deleted?
- If not, which function we should use to get the affected row?

```
$sql = "delete from MyTable where id=1";
```

```
try {
```

```
    mysqli_query($con, $sql);
```

```
    echo "Successfully deleted id=$id. " ;
```

```
}
```

```
catch (Exception $e) {
```

```
    echo "Error: " . $sql . "<br>" . mysqli_error($con);
```

```
}
```


Exercise – UPDATE in PHP programs

```
$result = mysqli_query($con, $sql);
```

- Explain what will be the value of the **\$result** if we have the following SQL in the PHP program?
 1. `$sql = "update MyTable set name=tester where id=1 ";`
 2. `$sql = "update MyTable set name='tester' where id=1 ";`
 3. `$sql = "update MyTable set name='tester' where id=0 ";`
 4. `$sql = "update MyTable set name='tester' where id=1 ";`
- Home exercises - Test the similar query for insert and delete operations, and check the results.

Class exercise in class DB: INSERT with FK and PHP #1

- If we have a table MyTable_xxxx (id, name, fid), where
 - id is auto increment primary key
 - fid is a foreign key
- What happens to the INSERT if the fid is null or empty? (see next 2 slides)

Class exercise in class DB: INSERT with FK and PHP #2

- **Drop and Re-create MyTable with FK key**

```
CREATE TABLE MyTable_xxxx(  
  id int auto_increment,  
  name varchar(20),  
  fid int,  
  Primary key(id),  
  Foreign key(fid) references dreamhome.Sources(id)  
);
```

- Reference

- https://www.w3schools.com/sql/sql_foreignkey.asp

Class exercise in class DB:

INSERT with FK and PHP #2

- What happens to the INSERT if the fid is null or empty? (Exercise)

(1) INSERT INTO MyTable_xxxx(fid,name)

VALUES (NULL,'John');

=> success, but the record has unknown parent (foreign key) – NULL value

(2) INSERT INTO MyTable_xxxx(fid,name)

VALUES ('','Nancy');

=> fail, check the foreign key constraint – no empty value allowed.

Class exercise in class DB: INSERT with FK and PHP #3

- What happens to the INSERT if the fid is null or empty?

```
$sql = "INSERT INTO MyTable_xxxx (name) VALUES ('John')";  
try {  
    mysqli_query($con, $sql);  
    $last_id = mysqli_insert_id($con);  
    echo "Inserted successfully. Last inserted ID is: " . $last_id;  
}  
catch (Exception $e) {  
    echo "Error: " . $sql . "<br>" . mysqli_error($con);  
}
```

Exercise - UPDATE with FK and PHP

- If we have a table MyTable_xxxx (id, name, fid), where
 - id is auto increment primary key
 - fid is a foreign key
- Based on these codes, can we tell if the record has been successfully updated?

```
$sql = "update MyTable set name='tester' where id=1";  
  
try {  
    mysqli_query($con, $sql);  
    echo "Successfully changed name to 'tester' for id=$id. " ;  
} catch (Exception $e) {  
    echo "Error: " . $sql . "<br>" . mysqli_error($con);  
}
```

DELETE with FK and PHP

- If we have a table MyTable_xxxx (id, name, fid), where
 - id is auto increment primary key
 - fid is a foreign key
- Based on these codes, can we tell if the record has been successfully deleted?

```
$sql = "delete from MyTable where id=1";
```

```
try {
```

```
    mysqli_query($con, $sql);  
    echo "Successfully deleted id=$id. " ;
```

```
}
```

```
catch (Exception $e) {  
    echo "Error: " . $sql . "<br>" . mysqli_error($con);  
}
```

Reading Assignment

- Familiar with PHP MySQLi functions that
- Execute the insert, update, delete operations
- Operations on tables that have auto increment and foreign keys relationship

Conclusion

- Work on project
- Write PHP programs to insert data into MySQL
 - Get data from HTML CGI inputs
 - Verify if the data is inserted into the table successfully
 - Print corresponding messages for success or fail
 - Get the last insert id and use it for other purposes